CS316

INTRODUCTION TO AI AND DATA SCIENCE

Topic 10

Hypothesis Testing

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Statistical Inference

What is Statistical Inference?

- Making generalizations about a population using data from a sample.
- Two main components:
  - **Parameter Estimation**
  - **Hypothesis Testing**

Term	Meaning
Parameter	A fixed value describing a population (e.g., population mean μ)
Statistic	A value calculated from sample data (e.g., sample mean $\vec{x}$)

Relevance of Statistical Inference to ML

- **You rarely have all the data**
  Inference helps you generalize from **samples to populations**
- **Machine Learning is built on probability**
  Every prediction is based on **estimated parameters**, uncertainty, and distributions
- **Model performance evaluation uses inference tools**
  Confidence intervals, p-values, hypothesis tests underlie:
  - Feature selection
  - Model comparison (e.g., is Model B *really* better than A?)
- **Avoid overfitting and underfitting**
  Inference helps validate whether patterns in your data are **real or spurious**
- **Interpretable AI depends on inference**
  Helps practitioners answer: *"How confident are we in this result?"*

Example

You predict that forwards are faster than defenders based on 200 player profiles.

Statistical inference answers:

- Is this difference **statistically significant**?
- Could it be due to **random variation**?
- What's the **margin of error** on your estimate?

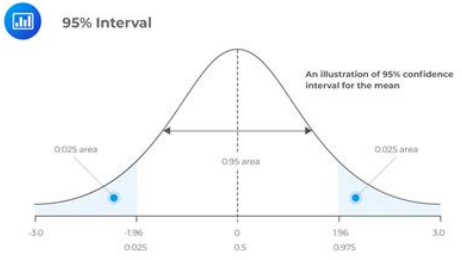
Parameter Estimation

Types of Estimation

Point Estimate: Single value (e.g., $\bar{x}$: sample mean)

Interval Estimate: Confidence Interval (e.g., 95% CI)

Confidence Interval (CI): A range likely to contain the true parameter. Example (95% CI):“We are 95% confident the true mean lies between 50 and 60.”



Confidence Intervals

Example: Estimate the average height of adult women using a sample.

Sample mean = 161.5 cm

95% CI = [160.2, 162.8]

Hypothesis Testing

Definition

A method for making decisions using data. It determines if observed patterns are real or just random chance.

Purpose

Provides a framework to systematically evaluate claims about populations using sample data.

Application

Used in research, medicine, business, and many scientific fields to test theories and make predictions.

Key Concepts

Term	Explanation
Null Hypothesis (H_o)	The default assumption - "nothing interesting is happening."
Alternative Hypothesis (H_1)	The claim we're testing for - usually that a difference exists.
p-value	Probability of observing our result if H_o is true.
Significance Level (α)	Threshold (commonly 0.05) for rejecting H_o.

T-test

1. One-Sample t-Test

Compares a sample mean ($\bar{x}$) to a known or hypothesized population mean (μ).

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

- $\bar{x}$ = Sample mean
- μ = Hypothesized population mean
- **s** = Sample standard deviation
- **n** = Sample size
- **Degrees of freedom (df)** = n - 1

https://numiqo.com/tutorial/t-distribution

```
from scipy import stats
t_stat, p_value = stats.ttest_1samp(data, popmean=70)
```

Use case:

Example: Checking a course's average exam score

A faculty claims that the average final exam score across all CS students is **70**. You take a random sample of **25 students** from one course and find:

- Sample mean ($\bar{x}$) = 74
- Sample standard deviation (s) = 8
- Sample size (n) = 25

You want to know whether this course's mean differs significantly from 70.

Step 1: Hypotheses

- H_0: $\mu = 70$ (the true mean equals 70)
- H_1: $\mu \neq 70$ (the true mean differs from 70)

Step 2: Test statistic

$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$

$t = (74 - 70) / (8 / \sqrt{25}) = 4 / 1.6 = \mathbf{2.5}$

Step 3: Degrees of freedom

$df = n - 1 = 24$

Step 4: Decision ($\alpha = 0.05$, two-tailed)

The critical value for df = 24 is about ±2.064. Since 2.5 > 2.064, we reject H_0. (The p-value is roughly 0.02, which is below 0.05.)

Conclusion: There is statistically significant evidence that the average score in this course differs from 70. It is higher, by about 4 points.

T-test

2. Independent Two-Sample t-Test (Equal Variances)

Compares the means of two independent groups assuming equal population variances.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where the **pooled standard deviation (s_p)** is:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

- $\bar{x}_1$, $\bar{x}_2$ = Means of group 1 and group 2
- s_1, s_2 = Standard deviations of group 1 and group 2
- n_1, n_2 = Sample sizes of group 1 and group 2
- **Degrees of freedom (df)** = $n_1 + n_2 - 2$

python

```
from scipy import stats

t_stat, p_value = stats.ttest_ind(group_a, group_b, equal_var=True)
```

Example: Comparing two teaching methods

A CS instructor wants to know whether students taught with a new active-learning approach (Group A) score differently on the final exam than students taught with traditional lectures (Group B). Students are in separate sections, so the groups are independent.

	Group A (Active learning)	Group B (Lecture)
Sample size	$n_1 = 20$	$n_2 = 22$
Sample mean	$\bar{x}_1 = 78$	$\bar{x}_2 = 72$
Sample std. dev.	$s_1 = 8$	$s_2 = 9$

Step 1: Hypotheses

- H_0: $\mu_1 = \mu_2$ (no difference in mean scores)
- H_1: $\mu_1 \neq \mu_2$ (the means differ)

Step 2: Pooled variance

$s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$

$s_p^2 = [(19)(64) + (21)(81)] / 40 = (1216 + 1701) / 40 = 2917 / 40 = \mathbf{72.93}$

$s_p = \sqrt{72.93} \approx \mathbf{8.54}$

Step 3: Test statistic

$t = (\bar{x}_1 - \bar{x}_2) / (s_p \cdot \sqrt{(1/n_1 + 1/n_2)})$

Standard error $= 8.54 \times \sqrt{(1/20 + 1/22)} = 8.54 \times \sqrt{0.0955} = 8.54 \times 0.309 \approx 2.64$

$t = (78 - 72) / 2.64 \approx \mathbf{2.27}$

Step 4: Degrees of freedom

$df = n_1 + n_2 - 2 = 40$

Step 5: Decision ($\alpha = 0.05$, two-tailed)

The critical value for df = 40 is about ±2.021. Since $2.27 > 2.021$, we reject H_0. (The p-value is roughly 0.03.)

Conclusion: There is statistically significant evidence that the two teaching methods produce different mean exam scores, with the active-learning group scoring about 6 points higher.

T-test

Paired Samples t-Test

Compares related pairs of observations (such as pre-test and post-test scores from the same group).

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

- $\bar{d}$ = Mean of the differences between paired values
- s_d = Standard deviation of the differences
- **n** = Number of pairs
- **Degrees of freedom (df)** = n - 1

https://numiqo.com/tutorial/t-distribution

Eight students take a diagnostic test before and after a workshop.

Student	Before	After	Difference (d = After − Before)
1	55	62	7
2	60	66	6
3	48	50	2
4	70	78	8
5	52	55	3
6	65	64	−1
7	58	66	8
8	62	69	7

Step 1: Hypotheses

- H_0: $\mu_d = 0$ (the workshop has no effect on average)
- H_1: $\mu_d \neq 0$

Step 2: Summary of differences

Mean difference: $\bar{d} = (7+6+2+8+3-1+8+7) / 8 = 40 / 8 = \mathbf{5}$

Standard deviation of differences: $s_d \approx \mathbf{3.30}$

Step 3: Test statistic

$t = \bar{d} / (s_d / \sqrt{n})$

$t = 5 / (3.30 / \sqrt{8}) = 5 / 1.167 \approx \mathbf{4.28}$

Step 4: Degrees of freedom

$df = n - 1 = 7$

Step 5: Decision ($\alpha = 0.05$, two-tailed)

The critical value for df = 7 is about ±2.365. Since $4.28 > 2.365$, we reject H_0 ($p \approx 0.004$).

Conclusion: There is significant evidence that scores changed after the workshop, with an average improvement of about 5 points.

Exercise

Test whether a classifier A is really better than a classifier B.

Run a t-test on 10-fold cross-validation results. (which type of t-test you would use ?)

Fold	1	2	3	4	5	6	7	8	9	10
Model A	0.82	0.80	0.84	0.79	0.83	0.81	0.85	0.80	0.82	0.81
Model B	0.85	0.83	0.84	0.82	0.86	0.82	0.88	0.81	0.85	0.83
Diff (B – A)	0.03	0.03	0.00	0.03	0.03	0.01	0.03	0.01	0.03	0.02

mean difference = $\frac{0.03+0.03+0+0.03+0.03+0.01+0.03+0.01+0.03+0.02}{10} = 0.022$

Exercise(solution)

Step 1: The data

Fold	1	2	3	4	5	6	7	8	9	10
Model A	0.82	0.80	0.84	0.79	0.83	0.81	0.85	0.80	0.82	0.81
Model B	0.85	0.83	0.84	0.82	0.86	0.82	0.88	0.81	0.85	0.83
Diff (B − A)	0.03	0.03	0.00	0.03	0.03	0.01	0.03	0.01	0.03	0.02

Average accuracy is 0.817 for A and 0.839 for B. B is better by **2.2 percentage points** in this sample, which is a statistic. The question is whether that gap is real or just luck of the folds.

Step 2: Hypotheses

- H_0: the true mean difference is zero ($\mu_diff = 0$) assume the error is low and due to randomness only
- H_1: $\mu_diff \neq 0$ (two-sided)
- $\alpha = 0.05$

Step 3: Compute the test statistic

- Mean difference: $\bar{d} = 0.22 / 10 = \mathbf{0.022}$
- Standard deviation of the differences: $s = \mathbf{0.0114}$
- Standard error: $SE = s / \sqrt{n} = 0.0114 / \sqrt{10} = \mathbf{0.00359}$
- t-statistic: $t = \bar{d} / SE = 0.022 / 0.00359 = \mathbf{6.13}$, with $df = n - 1 = 9$

Step 4: Decision

- The critical value for $df = 9$ at $\alpha = 0.05$ (two-sided) is **2.262**.
- Since $6.13 > 2.262$, we reject H_0. The p-value is about **0.0002**, far below 0.05.

Conclusion: Model B's improvement is statistically significant.

Step 5: Confidence interval

95% CI = $0.022 \pm 2.262 \times 0.00359 = 0.022 \pm 0.0081$, so **[0.014, 0.030]**.

We are 95% confident that B's true advantage is between **1.4 and 3.0 percentage points**. The interval excludes 0, which agrees with the test.

CI = estimate ± (critical value) × (standard error)

$\bar{d} \pm \text{threshold} \times s_d / \sqrt{n}$

Type I and Type II Errors

	Type I Error (False Positive)	**Type II Error (False Negative)**
Definition	Rejecting the null hypothesis when it is actually **true**	Failing to reject the null hypothesis when it is actually **false**
Notation	**α** (alpha) – Significance level	**β** (beta) – Probability of missing a real effect
What You're Saying	"There **is** an effect" (but there isn't)	"There **is no** effect" (but there is one)
Real-World Example	A **medical test** says a healthy person has a disease (false alarm)	A **medical test** misses a real disease in a sick person (missed case)
Consequence	False discovery, overreaction	Missed opportunity, underreaction
How to Reduce	Lower the **significance level (α)** (e.g., from 0.05 to 0.01)	Increase **sample size** or use more powerful tests
When It's Tolerated	In **exploratory research** or **non-critical decisions**	In **life-critical systems** (e.g., medicine)

SUMMARY

We have explored the concepts of:

- Hypotheses and the null hypothesis.
- Statistical inference.
- Statistical testing.

Next, AI and Data Science Ethics.